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Why do we need statistical models?

- how netflix know which movies you might like next?

  data ⟹ statistical model ⟹ recommendation

- for the same ques, how does GPT gives diff. ans. everytime?

  prompt ⟹ probab. distribuⁿ over vocabulary ⟹ sampling ⟹ output

- for a particular incident, why did my credit card got declined?

  normal spending pattern ⟹ transacaⁿ is unusual ⟹ most likely fraud ⟹ card declined

- all of such problems begins with data, which is generally large, noisy & heterogenous.
- we try to learn the statistical relationships btw. the data pts., rather than remembering past pts.
- once the model has learned, we can make predictions, estimate probabilities, or rank alternatives
- since the predictions have error, we must have a method to quantify this error. And showcase how our model will generalize.
- statistical skill we must have -
  - → understand data & its limitaⁿˢ
  - → build statistical models
  - → estimate / predict unknown quantities.
  - → quantify errors in the predictions
  - → compete the models fairly, understant when or not the models be trusted.
- why the statisticians do not trust intuition?

human intuition is unreliable when reasoning under uncertainity, on the other hand, Statistics gives you a principled framework. Initially, we start with assumptions, how the

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data was generated.

- when we have statistical thinking, we ignore about the fact that is or is not the story is convincing.

$$\underbrace{P(A \cap B) \leq P(A)}$$

probability determines the answer not plausibility.

- Observa^n → it's outcome is unknown. Different outcomes are possible.
- Statistics is the science of reasoning under uncertainity.
-

Deterministic		Stochastic	
same input →	same output	same input →	any outcome

- statistics becomes necessary when variability is the part of the system.
- Observed variability can have multiple sources. Using statistics, we try to understand & quantify them.
- Observable data = noise + signal

Signal → systematic component that reflects the underlying phenomenon.

Noise → is that variation which can't be explained with the help of variables under study.

- We can't ignore both. Noise helps us to discover patterns, while signal helps us to find crucial scientific relationships.
- In statistics, we try to extract signal while also accounting for noise.

Two Types - Aleatoric Uncertainity → The uncertainity that will continue to exist even if we know the best possible statistical model & info. It shows

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the inherent variability of the data-generating mechanism.

- Epistemic uncertainity - This arises from the incomplete knowledge about data generation. This can be avoided by getting more data, measuring more variables, improving models.
- Error decomposition is a property of nature, observer, and model.
- Often, better statistical models convert randomness into explainable variation.
- If uncertainity is -

  more epistemic
  - → collect data
  - → improve model
  - → measure more variables.

  more aleatoric
  - → estimate variability
  - → quantify uncertainity
  - → recognize predicⁿ limits.
- In statistics, we move from observed data to data-generating mechanism, and finally scientific phenomenon.
- smaller samples $\xRightarrow[\text{lead to}]{}$ more variability
- Law of small nos. → small samples of a populaⁿ fluctuate more than large samples of the same populaⁿ.